Support-Conditioned Evaluation Auditing for Aerial Vehicle Detection

Xinling Wen, Jiahao Zhang, Wenqi Liu, Yangxu Ren, and Yu Chen

***Abstract*—Aerial-detection scores can change because a small-object rule changes the annotation support being scored. We formulate support-conditioned evaluation as one chain: annotation-support mismatch, controlled support perturbation, metric uncertainty, and rank robustness on common coverage. The support layer reports retained valid ground truth under absolute and normalized rules. The perturbation layer separates target-set removal from source-ignore treatment and reports a policy band. The rank layer compares frozen configuration pools only on their shared image set and pairs margins with image-paired bootstrap intervals. At 24 pixels, retained vehicle support is 73.3%, 95.1%, and 9.7% for VisDrone, UAVDT, and AI-TOD; at normalized .015 it is 89.8%, 97.4%, and 81.0%. For one registered baseline artifact per source, the corresponding F_1 policy bands are .036/.025, .009/.002, and .330/.136; the AI-TOD values are conditioned on its 1,869 covered images. In a five-configuration, 548-image VisDrone pool, headline Y11m–ASD margins are .0449 [.0391, .0506] and .0277 [.0234, .0321], whereas observed rank-boundary intervals span zero. The audit distinguishes a changed target, a policy-sensitive metric, and an uncertainty-qualified within-pool comparison.**

***Index Terms*—Aerial detection, annotation support, evaluation sensitivity, rank robustness, vehicle detection.**

I. Introduction

A reported detection score is meaningful only after its scored target is fixed. In aerial imagery, a support rule can change that target sharply: the same pixel cutoff may retain most vehicles in one source while removing most vehicles in another. Resolution, ontology, source-provided ignores, and tiny-object conventions therefore enter an evaluation result before one compares detector outputs. This issue is consequential for heterogeneous aerial benchmarks [1], [2], [3], [4] and complements research on dataset bias, label quality, and error diagnosis [5], [6], [7].

Recent GRSL studies illustrate both the importance of high-resolution remote-sensing processing and the sensitivity of validation to geolocation conditions [8], [9]. What remains underreported in cross-artifact detection comparisons is a connected account of *what annotation support changed*, *how the metric reacts to that change*, and *whether the resulting model conclusion holds on a common denominator*. Reporting many evaluation switches without this ordering obscures the scientific question.

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Several adjacent lines motivate this account. Unified label spaces address semantic compatibility across datasets [10]; slicing and other tiny-object procedures change the prediction process itself [11]; noise-aware evaluation and label-error screening study annotation reliability [12], [13], [14]; and error taxonomies localize detector failure modes [7]. Our focus begins after an artifact and prediction cache are fixed: it records how an explicit change in scored support propagates into a metric and then into a within-pool decision.

We introduce a support-conditioned audit with the following three layers.

1) *Annotation-support mismatch:* report raw prediction coverage and retained valid ground truth under absolute and normalized support rules.
2) *Controlled support perturbation:* hold prediction artifacts fixed, vary the target/ignore contract explicitly, and summarize the resulting metric uncertainty by a policy band.
3) *Rank robustness on common coverage:* compare a frozen configuration pool only on shared images, map finite-grid winner regions, and pair a point margin with image-paired uncertainty.

Together, the three layers turn a reported score into a traceable support-to-decision record.

II. Support-Conditioned Audit

A. Artifacts, mapping, and coverage

We use VisDrone2019-DET validation, the UAVDT test split represented in COCO format for the common evaluator, and AI-TOD validation [1], [2], [3]. The available AI-TOD crop reconstruction has xView-derived provenance [15]. Vehicles are mapped as VisDrone {car, van, truck, bus}, UAVDT {car, truck, bus}, and AI-TOD {vehicle}. Raw coverage is determined from image names in a prediction artifact *before* class filtering; an executed image with no retained vehicle prediction is consequently evaluated as an empty prediction. Table I gives the support denominator. DOTA-v2 quadrilateral-to-HBB scale evidence is retained as a structural supplementary check, not as a metric or rank result [4].

The metric layer replays one registered baseline artifact per source. Its VisDrone and UAVDT records have complete image coverage; the AI-TOD record is explicitly conditioned on the 1,869 covered images. The rank layer is narrower: its primary VisDrone pool comprises five fixed configurations—two resolutions of one baseline checkpoint (B-640 and B-1280),

TABLE I
SUPPORT DENOMINATOR AND RETAINED VEHICLE GROUND TRUTH. COVERAGE IS RAW-PREDICTION IMAGE COVERAGE. ρ_{24} AND $\rho_{.015}$ ARE RETAINED FRACTIONS AT THE TWO HEADLINE RULES.

Source	Coverage	Vehicle boxes	ρ_{24}	$\rho_{.015}$
VisDrone	548/548	17,040	73.3%	89.8%
UAVDT	305/305	11,054	95.1%	97.4%
AI-TOD	1,869/2,804	59,904	9.7%	81.0%

independently trained Y11m and ASD configurations, and a fine-tuned Y11n-P2 configuration—each with 548/548 raw-image coverage. UAVDT remains an ancillary full-coverage ranking control; AI-TOD is excluded from rank claims.

B. Support and perturbation contract

For a ground-truth box (w, h) in an image (W, H), define absolute and normalized support as

$$s_a = \max(w, h), \qquad s_n = \max\left(\frac{w}{W}, \frac{h}{H}\right). \tag{1}$$

For source s and rule t, $G_s(t)$ denotes valid mapped boxes retained by the rule, and $\rho_s(t) = |G_s(t)|/|G_s|$ is the retained-support fraction. We register $s_a \in \{16, 20, 24, 28, 32, 40, 48, 64\}$ pixels and $s_n \in \{.005, .0075, .010, .015, .020, .030\}$; 24 pixels and .015 are the headline interior values.

We define three primary policies. I retains all mapped valid boxes. O removes below-support valid boxes and disables source-ignore neutralization. N uses the same filtered target as O, while enabling source ignores after valid-GT matching. Under O and N, a prediction overlapping a removed valid box remains eligible as a false positive; the separate removed-as-ignore control is supplementary. For a fixed support rule and metric m, the fixed-rule policy band is

$$B_m^{\pi}(t) = \max_{\pi \in \{I,O,N\}} m(t, \pi) - \min_{\pi \in \{I,O,N\}} m(t, \pi). \tag{2}$$

We use global micro-F_1 at confidence .25 and IoU .25, with score-ordered, valid-GT-first greedy matching. Source-ignore neutralization uses the COCO crowd overlap convention: an otherwise unmatched prediction is suppressed when its intersection-over-prediction-area is at least .5 [16]. AP-cap and matching controls are reported separately in the supplementary material.

C. Common-coverage rank protocol

For two configurations a, b and a registered rule r, the pairwise margin is $\Delta_{ab}(r) = m_a(r) - m_b(r)$. We evaluate margins only on the common image set and treat exact ties as unstable. The headline comparison fixes policy N and IoU .25. A supplementary 14-threshold-by-eight-confidence sensitivity surface locates finite-grid decision boundaries; it does not extrapolate beyond the declared grid. For headline and observed-boundary cells, a paired bootstrap resamples the 548 images jointly for both configurations 10,000 times and reports the percentile 95% interval of Δ_{ab}.

D. Formal properties of the audit

The two support coordinates deliberately answer different questions. Under isotropic resizing by a factor $\alpha > 0$, absolute support becomes $s'_a = \alpha s_a$, whereas

$$s'_n = \max\left(\frac{\alpha w}{\alpha W}, \frac{\alpha h}{\alpha H}\right) = s_n. \tag{3}$$

Thus normalized support is invariant to isotropic image resizing, while a fixed-pixel rule remains tied to the delivered image scale. This property does not make either coordinate universally preferable; it explains why their retained-target fractions must be reported separately.

Common coverage is an identifiability condition rather than a cosmetic balancing choice. If two configurations are observed on different image subsets and their TP, FP, and FN contributions on unobserved images are unconstrained, the observed-subset counts cannot identify their full-population micro-F_1 ordering: one completion of the missing images can favor a, while another can favor b without changing any observed count. We therefore make no rank statement for incompletely covered AI-TOD.

The deterministic and resampling parts of the rank audit are also distinct. For a finite audited rule set $\mathcal{R}_h$, define the candidate-level rule band and the reference-winner differential radius as

$$B_k(\mathcal{R}_h) = \max_{r \in \mathcal{R}_h} m_k(r) - \min_{r \in \mathcal{R}_h} m_k(r), \tag{4}$$

$$\Gamma_j(\mathcal{R}_h) = \max_{r \in \mathcal{R}_h} |\Delta_j(r) - \Delta_j(r_0)|, \tag{5}$$

where $k^\star$ wins at r_0 and $\Delta_j(r) = m_{k^\star}(r) - m_j(r)$. If $\Delta_j(r_0) > \Gamma_j(\mathcal{R}_h)$ for every competitor j, then $k^\star$ remains the winner throughout $\mathcal{R}_h$; moreover,

$$\Gamma_j(\mathcal{R}_h) \leq B_{k^\star}(\mathcal{R}_h) + B_j(\mathcal{R}_h). \tag{6}$$

The certificate concerns a declared finite rule set. The paired bootstrap instead asks whether a selected pairwise advantage is resolved under image resampling. A positive deterministic margin therefore does not substitute for an interval, and a wide candidate-level rule band does not by itself imply a rank reversal.

III. RESULTS

A. Layer 1: Annotation-support mismatch

Figure 1 is the first link in the chain. A common 24-pixel threshold changes the retained target from nearly all UAVDT vehicles to fewer than one tenth of AI-TOD vehicles. The .015 normalized rule narrows this disparity, retaining 81.0–97.4% across the three sources at the headline value. The result is not that one support domain is universally preferable; it is that the target shift is measurable and should accompany the score that follows it.

The coverage record also decides which sources can progress to the later layers. AI-TOD's covered subset contains 16,900 vehicle boxes in 1,869 images (9.0 boxes/image), whereas its 935 noncovered images contain 43,004 boxes (46.0 boxes/image). We therefore retain AI-TOD for support diagnosis rather than a model comparison. The complete VisDrone and

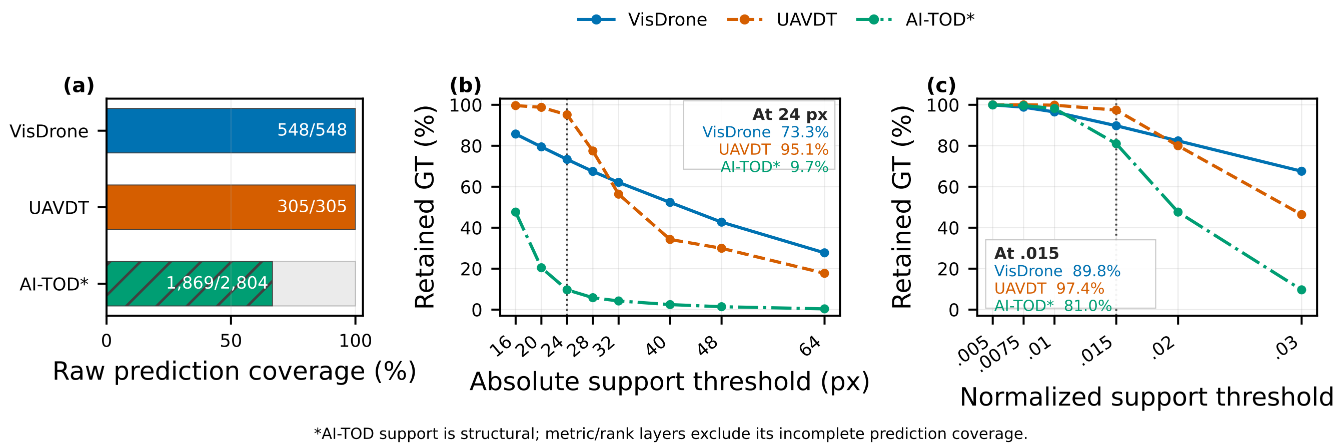

Fig. 1. Annotation-support mismatch. (a) Raw prediction coverage establishes the evaluation denominator before vehicle-class filtering. (b) A 24-pixel cutoff retains 73.3%, 95.1%, and 9.7% of vehicle ground truth in VisDrone, UAVDT, and AI-TOD. (c) The normalized .015 rule retains 89.8%, 97.4%, and 81.0%. AI-TOD is displayed for support diagnosis only because its prediction coverage is incomplete.

UAVDT records, by contrast, preserve images with empty retained predictions and their false-negative contribution.

B. Layer 2: Controlled perturbation across three sources

Table II traces the support mismatch into the metric layer under one baseline artifact per source. At 24 pixels, the policy band is .036 for VisDrone, .009 for UAVDT, and .330 for coverage-conditioned AI-TOD; at .015, the corresponding bands are .025, .002, and .136. The same nominal rule therefore produces not only different retained targets but also markedly different metric responses. AI-TOD's covered subset is reported rather than silently treated as the full validation set.

Within the five-configuration VisDrone pool, Fig. 2 shows that 24-pixel bands range from .031 (Y11m) to .093 (B-1280); at .015, they range from .013 (Y11n-P2) to .035 (Y11m). Filtering can raise or lower a score because it changes the valid target and the treatment of unmatched predictions. The figure therefore reports uncertainty of the evaluation contract, rather than attributing a policy-sensitive score to an unobserved annotation-error mechanism.

The O–N comparison isolates source-ignore handling at the same filtered target. At 24 pixels, enabling valid-first source-ignore neutralization moves F_1 by .008–.022 across the five configurations. The larger I–filtered separations in Fig. 2 show why the support rule and removed-object treatment must be stated together: a single endpoint alone cannot reveal which evaluative change produced it.

For the VisDrone B-640 artifact, N is below I at 24 pixels but above I at .015. Because predictions are unchanged, this sign change cannot be interpreted as detector improvement or degradation. UAVDT and AI-TOD contain no active source-ignore regions in this mapping, so $O = N$ there; their bands isolate target filtering. The nonzero VisDrone O–N separation isolates source-ignore neutralization at the same filtered target.

C. Layer 3: Rank robustness on common coverage

The headline comparison is made on exactly the same 548 images for all five configurations. At absolute 24 pixels, Y11m has $F_1 = .8181$ versus .7732 for ASD, giving $\Delta = .0449$ with a paired 95% interval [.0391,.0506]. At normalized .015, the corresponding values are .8741 and .8465, with $\Delta = .0277$ [.0234,.0321]. Y11m remains the winner and ASD the runner-up at both declared headline rules; the lower B-640/B-1280 order changes, which Fig. 3(a) makes visible rather than averaging away.

Supplementary sensitivity surfaces identify where a point ordering is not an established advantage. At the normalized .010/confidence-.30 boundary, Y11m–ASD is .00015 [$-$.00435, .00477]; at the absolute-64-pixel/confidence-.35 boundary, B-640–Y11m is .00095 [$-$.00492, .00677]. Table III records the four paired comparisons. The conclusion is therefore scoped to the declared headline rules: their Y11m–ASD margins are resolved on common coverage, whereas near-boundary ranks require an uncertainty qualifier.

This contrast separates score movement from a model conclusion. A fixed rule can move several configurations in the same direction while preserving their pairwise separation. Conversely, a small differential response can overturn a near tie. The sensitivity surface locates these finite-grid decision regions; the paired interval determines whether a selected pairwise advantage is resolved at the image level.

IV. Discussion and Reproducibility

The audit turns an evaluation question into a traceable sequence. First report how much annotation support remains; then expose how an explicit target/ignore perturbation moves the metric; finally state whether a frozen configuration-pool conclusion persists on common coverage and under paired uncertainty. This sequence separates target drift from metric sensitivity and from a model comparison.

The three numerical objects have different estimands. $\rho_s(t)$ is a pooled box-level property of the mapped annotations and is computed before prediction matching. $B^{\pi}_{m}(t)$ is a deterministic range over three declared evaluator policies for one fixed prediction artifact; it is not a confidence interval. $\Delta_{ab}(r)$ is a pairwise difference on a named common image set, while its

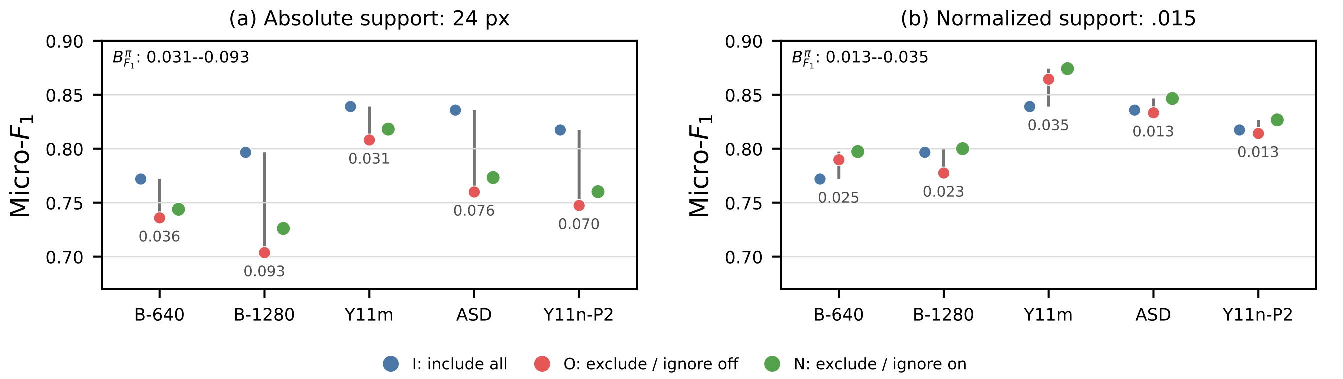

Fig. 2. Controlled support perturbation and metric uncertainty on the five-configuration VisDrone pool. Each vertical span is $B^{\pi}_{F_1}(t)$ for one configuration; colored points are I (blue), O (red), and N (green). (a) Absolute-24-pixel bands span .031–.093. (b) Normalized-.015 bands span .013–.035.

TABLE II

THREE-SOURCE BASELINE REPLAY AT THE HEADLINE RULES. ρ_{cov} IS RETAINED GT ON THE METRIC-COVERED IMAGES; CONFIDENCE AND IOU ARE .25.

Source	Candidate	Coverage	Rule	ρ_{cov}	I	O	N	$B^{\pi}_{F_1}$
VisDrone	B-640	548/548	24 px	73.3%	.7719	.7359	.7438	.0360
			.015	89.8%	.7719	.7897	.7973	.0254
UAVDT	UAVDT-Base	305/305	24 px	95.1%	.3453	.3367	.3367	.0085
			.015	97.4%	.3453	.3430	.3430	.0023
AI-TOD†	AI-TOD-Base	1,869/2,804	24 px	12.0%	.3952	.0648	.0648	.3303
			.015	55.3%	.3952	.2593	.2593	.1359

†AI-TOD endpoints and ρ_{cov} are conditioned on the 1,869 images with raw prediction coverage; the source-level support fractions over all 2,804 images are 9.7% and 81.0% in Table I.

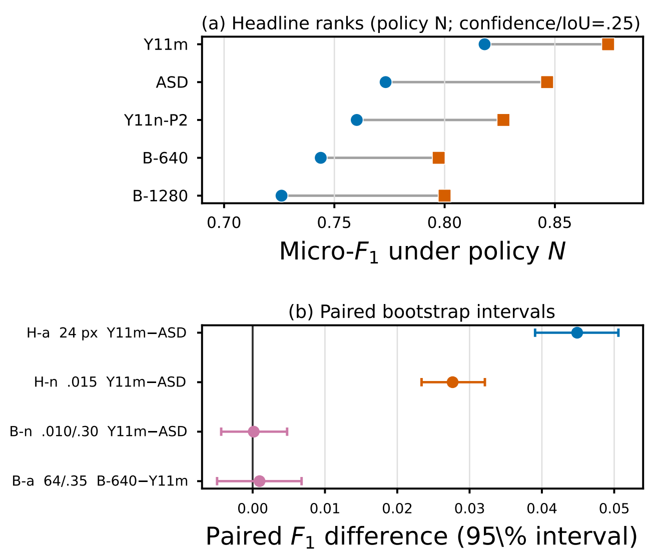

Fig. 3. Common-coverage rank robustness for five fixed VisDrone configurations. (a) Blue circles are absolute-24-pixel and orange squares normalized-.015 F_1 values under policy N; all configurations use the same 548 images. (b) H-a/H-n are the two headline comparisons and B-a/B-n are observed finite-grid boundary cells. Error bars are 95% image-paired bootstrap intervals from 10,000 resamples.

bootstrap interval uses images, not boxes, as the resampling unit. Reporting them in one record prevents a large support shift from being described as statistical uncertainty and prevents

TABLE III

COMMON-COVERAGE PAIRED F_1 RECORDS (10,000 IMAGE-PAIRED REPLICATES). H = HEADLINE; B = OBSERVED FINITE-GRID BOUNDARY.

Cell	Pair	Δ	95% interval
H-a: 24 px, .25	Y11m–ASD	.0449	[.0391,.0506]
H-n: .015, .25	Y11m–ASD	.0277	[.0234,.0321]
B-n: .010, .30	Y11m–ASD	.00015	[−.00435,.00477]
B-a: 64 px, .35	B-640–Y11m	.00095	[−.00492,.00677]

an image-level interval from being generalized to unobserved images or configurations.

This separation yields an operational interpretation. First, a result is *target-sensitive* when its retained-support denominator changes materially. Second, it is *contract-sensitive* when the policy band is non-negligible relative to the scientific comparison. Third, a model conclusion is *resolved on common coverage* only when the candidate pair uses the same image universe and the paired interval supports the stated direction. These labels are cumulative: a comparison may be target- and contract-sensitive while still having a resolved pairwise margin, as the two headline Y11m–ASD records demonstrate.

For a compact comparison record, the three layers translate into three released objects: a source-level retained-support curve with the raw coverage denominator; a candidate-level table of policy endpoints and $B^{\pi}_{m}(t)$; and a pairwise common-coverage record containing the rule, point margin, resampling unit, and interval. These objects can accompany a conventional AP or

TABLE IV
MINIMAL SUPPORT-CONDITIONED COMPARISON RECORD.

Layer	Released primitive and question resolved
Support	Raw image coverage, $\rho_s(t)$, and the support coordinate: *which valid target is scored?*
Perturbation	$I/O/N$ endpoints and $B_m^{\pi}(t)$: *how much does the metric move under the declared target/ignore contract?*
Rank	Common image set, pairwise Δ_{ab}, resampling unit, and interval: *is a pool-specific ordering resolved?*

F_1 table without altering a detector. They identify whether an apparent gain follows a changed target, a changed contract, or a reproducible candidate difference.

The record is deliberately ordered. Support can be measured from annotations before predictions are replayed; metric uncertainty then holds a prediction artifact fixed while the contract changes; rank evidence is finally meaningful only after the compared configurations share the same image universe. This ordering gives a reviewer or downstream user a direct path from a plotted score to the target and comparison on which it rests.

The scope follows directly from those denominators. AI-TOD establishes both a support mismatch and a coverage-conditioned metric response, but not a rank comparison: 935 of 2,804 images lack raw prediction coverage and the missing subset is structurally denser than the covered subset. The five VisDrone configurations support a within-pool conclusion on 548 shared images, not a detector-family leaderboard. Likewise, the finite threshold grid locates observed decision boundaries but does not certify rules outside that grid. Widening the claim would require new common-coverage artifacts or an explicit model for the missing outcomes.

The supplementary material contains the complementary controls: full sensitivity surfaces, greedy–Hungarian matching checks, AP cap-contract checks, and structural DOTA-v2 evidence. These controls test evaluator boundaries without enlarging the central common-coverage rank claim. Code, configurations, derived evaluation records, figure-source tables, and checksums are available at https://github.com/Marchematics/aerial-evaluation-audit; licensed imagery and raw prediction artifacts remain referenced by manifest and hash.

V. CONCLUSION

Support-conditioned auditing connects annotation-support mismatch to a controlled metric perturbation and an uncertainty-qualified rank result. Across three declared aerial sources, the chain exposes large target shifts and source-dependent policy bands; on the common-coverage VisDrone pool, it further separates robust headline margins from unresolved boundary rankings. The practical recommendation is to release the support denominator, perturbation contract, and common-coverage paired comparison with each aerial-detection result.

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